

Master IARFID

Reconocimiento de Escritura (RES)

Practical session 4:

Indexing & Search on Handwritten Text Images

PRHLT-Group



Departamento de Sistemas Informáticos y Computación
Universitat Politècnica de València



KWS Approach

The KWS approach for indexing handwritten documents has the following features:

- ▶ Lexicon-based approach.
- ▶ Query set composed of single words.
- ▶ Line images are the basic spotting units; that is, we try to answer if a given word query is or is not in a line image.
- ▶ Word confidence score computation are based on the line level word graphs generated previously from the Rodrigo dataset.
- ▶ Performance evaluation based on *precision-recall* measures.

Most recent publication: <http://www.mdpi.com/2313-433X/4/1/15>

Required Tools

The following software will be used:

- **WordGraph2Index**: to produce probabilistic index from a word graph

```
mkdir -p Exp-KWS/WGIDX;  
cd Exp-KWS  
  
git clone https://github.com/PRHLT/WordGraph2Index.git  
cd WordGraph2Index  
make;  
cd ..
```

- **kws-assessment**: to measure the spotting performance

```
git clone https://github.com/PRHLT/KwsEvalTool.git  
cd KwsEvalTool  
gcc -Wall -O4 -o kws-assessment kws-assessment.c;  
cd ../..
```

Query Selection and Indexing

1. Selecting all keywords from transcripts with length greater than 1:

```
gawk '{if (length($1) > 1 && $1 !~ /[<#]/ ) print $1}' \
models/WFST/wordsMap.txt > Exp-KWS/keywords.lst
```

2. Obtaining index files from word graphs applying *posteriorgram* confidence score approaches.

```
for f in results/lattices/words/*.lat.gz; do
  F=$(basename $f .lat.gz); echo "Processing $F ..."; \

  Exp-KWS/WordGraph2Index/WordGraph2Index -i $f -z w | \
  gawk '!/^#|NULL/{print $2,$3,$4,$5,$6}' > \
  Exp-KWS/WGIDX/${F}_p.idx; \
done
```

Building Index File for Evaluation

3. From each word in the word graph mark if appears on the line transcription.
And generated entries of the form:

< lineId	word	[0 1 -1]	prob >
----------	------	----------	--------

- 0 means the word on the word graph do not appear in the line ground truth.
- 1 means the word on the word graph appears in the line ground truth.
- -1 means that this word do not appears on the word graph but in the ground truth.

```
for file in `cat data/lists/test.lst`; do
    [ -e Exp-KWS/WGIDX/${file}_p.idx ] && \
    echo Exp-KWS/WGIDX/${file}_p.idx;
done > Exp-KWS/indx.lst
```

Building Index File for Evaluation

```
gawk -v TR=data/text/test_words.ref ' BEGIN{
    while ((getline < TR) > 0)
        for (w=2; w<=NF; w++) REF[$1][$w]=1;
    }{
        lineId=$1;
        gsub("_p.idx", "",lineId)
        gsub("./", "",lineId)
        while ((getline word_line < $1)>0) {
            split(word_line, word);
            words_in_WG[word[1]]=1;
            result=0;
            if (word[1] in REF[lineId])
                result=1;
            print lineId,word[1],result,word[2]
        }
        for (w in REF[lineId]) #Words in REF but not in the WordGraph
            if (!(w in words_in_WG)){ print lineId,w,"1","-1"}
        delete words_in_WG;
    }' Exp-KWS/indx.lst > Exp-KWS/IDX_p.dat
```

Performance Search Evaluation

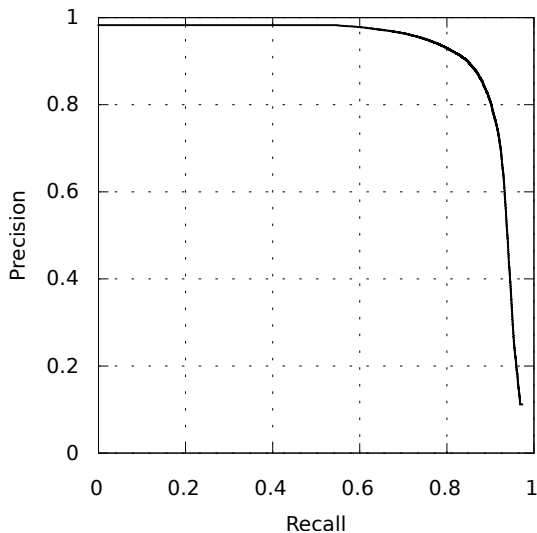
4. Computing **AP** and **mAP**:

```
N_WG=`wc -l Exp-KWS/indx.lst|cut -d ' ' -f 1`  
# ----> 3359 lines  
  
./Exp-KWS/KwsEvalTool/kws-assessment \  
-t -s -a -m Exp-KWS/IDX_p.dat -w Exp-KWS/keywords.lst -l ${N_WG}  
# MAP = 0.897 ( #Rel-Wrds = 2792 )  
# AP = 0.919
```

5. Computing and drawing **R-P** curve:

```
./Exp-KWS/KwsEvalTool/kws-assessment -t -s Exp-KWS/IDX_p.dat \  
-w Exp-KWS/keywords.lst -l 5011 > Exp-KWS/r_p_data.dat  
  
cp ./Exp-KWS/KwsEvalTool/egs/plot-R-P.gnp Exp-KWS;  
cd Exp-KWS/  
gnuplot plot-R-P.gnp  
evince R-P.pdf  
cd -
```

Evaluation



The graphical query tool

- From line index files to page index files, adding line info

```
mkdir Exp-KWS/WGIDX/pages

for file in `cat data/lists/corpus.lst`;
do
    NOM=`basename $file`;
    [ -e Exp-KWS/WGIDX/pages/${NOM}.idx ] && \
    rm Exp-KWS/WGIDX/pages/${NOM}.idx

    echo $NOM;
    for idx in `ls Exp-KWS/WGIDX/${NOM}*.idx`;
    do
        ID_LINE=`basename $idx .idx|sed -e "s/${NOM}_//" -e s"/_p$/"`
        awk -v nom=$ID_LINE '{print $0, nom}' $idx >> \
        Exp-KWS/WGIDX/pages/${NOM}.idx
    done
done
```

The graphical query tool

- ▶ Download the graphical query tool.

```
source res_env/bin/activate
git clone https://github.com/moisesPastor/GPWS.git
pip install PyQt5
```

- ▶ Scale word bounding boxes to page height

```
cd GPWS/EscalaIndex; make; cd -

mkdir Exp-KWS/WGIDX/pages_esc
for file in Exp-KWS/WGIDX/pages/*.idx; do
    echo $file;
    NOM=`basename $file .idx`

    GPWS/EscalaIndex/escalaIndex -i $file \
        -x data/Corpus_clean/${NOM}.xml \
        -o Exp-KWS/WGIDX/pages_esc/${NOM}.idx
done

ls Exp-KWS/WGIDX/pages_esc/*.idx | sed "s/.idx//" > pages.lst
```

The graphical query tool

► Run the query tool

python GPWS/main.py pages.lst data/Corpus Exp-KWS/WGIDX/pages_esc

Handwritten Page Image Word Searcher (on carxot)

Palabra

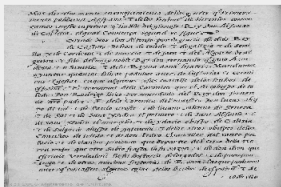
Probabilidad

Buscar

Buscador de palabras en imágenes de páginas

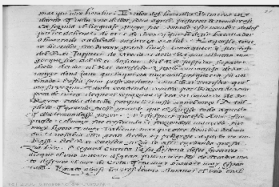
40 páginas encontradas

Imágenes



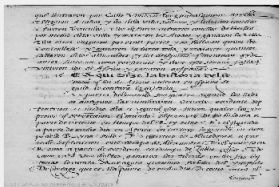
1.00

Rodrigo_00035.jpg



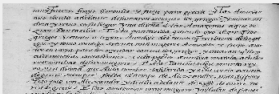
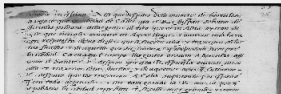
1.00

Rodrigo_00048.jpg



1.00

Rodrigo_00075.jpg



The graphical query tool

Imagen: Rodrigo_00048.jpg

Prob: 1.00 (on carxot)

mas que otro hombre. E alos dos hercules Primeros cuy
dando q cada vno dellas seaia aquel, pusieron le nombre asi
Ca segund el lenguaje griego fue tomado este nombre de dos
partes de letras, de es y de cloos q quizee dezir batalla dor
o honrrado o alabado en fuerza o en lid. E a queste terre
ro hercules fue de muy grand linaje como quier q fue hijo
del Rei Juppiter de Grecia y de la Reyna almena mu
ger que fue del Rei Anfiton. El Rei Juppiter supadre
diolo Acirar al Rei euristelo y fizolo con consejo de su
muger donã Juno que lo quezia muy mal porque era su en
tenado. Esta Juno puso su Amor con el Rei Euristelo que
en su vezino. Y aua contienda con ella por Razon de vna
pora de tierra de que el se pagaua q era en comarca de su
Reyno. Ella diosela porque le criasse aqael mogo. A tal
pleito q quando fuesse grande que el fiziesse todo a quello

Escala

Anterior

Siguiente

Cerrar

3 páginas

00

00075.jpg